Drawing Windows

Introduction

In this unit we are going to cover how to build drawing windows.

JFrame Class:

The JFrame class is used to create a window. We will extend this class to create our windows. Below is an overview of the JFrame class.

Constructors:

Constructor	Description
JFrame()	Creates a frame with no title
JFrame(String name)	Creates a frame with a title equal receive name (String)

Methods:

Return Type	Name	Description
void	setSize(int width, int height)	Sets the size of the window to the given width and height
void	setPreferredSize (Dimension d)	Sets the preferred size of the window or sets it back to its defaults if null is sent.
Insets	getInsets()	Returns the insets of the container. The insets are the space used by the title bar and borders.
int	getWidth()	Returns the frame's width.
int	getHeight()	Returns the frame's height.
void	pack()	Causes the window to be sized based on its preferred size.
void	setVisible(boolean show)	Makes the window hidden when show is false or visible when show is true
void	setLayout(LayoutManager manager)	Sets the window's layout manager to the received manager. If null is received no layout manager is used.
void	add(Component c)	Adds the given component to the frames content pane.
void	remove(Component c)	Removes the given component from the frames content pane.
void	paint(Graphics g)	Paints the container and its components.
void	setDefaultCloseOperation(int operation)	This method is used to set what happens when the x of your frame is clicked. We will always set it to JFrame.EXIT_ON_CLOSE. This value will make the x close down your program.
void	setUndecorated (Boolean hide)	When hide is true the frame's boarder and title bar will be hidden, they will be visible when hide is false

JPanel Class:

The JPanel class is a class that works as a canvas for drawing. We will extend this class to create a drawing area that we can add to our JFrame. Below is an overview of the JPanel class.

Constructors:

Constructor	Description
JPanel()	Creates a panel

Methods:

Return Type	Name	Description
void	setSize(int width, int height)	Sets the size of the window to the given width and height
int	getWidth()	Returns the frame's width.
int	getHeight()	Returns the frame's height.
void	setLayout(LayoutManager manager)	Sets the window's layout manager to the received manager. If null is received no layout manager is used.
void	add(Component c)	Adds the given component to the frames content pane.
void	remove(Component c)	Removes the given component from the frames content pane.
void	paint(Graphics g)	Paints the container and its components.

Graphics & Drawing

To draw to a JPanel override the paint method and then use the Graphics object to draw to the window. Below are the Graphics commands we will use for drawing.

Methods:

Return Type	Name	Description
void	setColor(Color c)	Changes the drawing color to c.
void	drawArc(int x, int y, int width, int height, int startAngle, int arcAngle)	Draws the outline of a circular or elliptical arc covering the specified rectangle.
void	drawLine(int x1, int y1, int x2, int y2)	Draws a line from (x1,y1) to (x2,y2)
void	drawOval(int x, int y, int width, int height)	Draws the outline of an oval inscribed in an invisible rectangle with a corner at (x,y) and covers the given width/height.
void	drawPolygon(int[] xPoints, int[] yPoints, int nPoints)	Draws the outline of a polygon with points taken from the xPoints and yPoints arrays. nPoints is the length of the arrays.
void	drawPolygon(Polygon p)	Draws the outline of polygon p.
void	drawPolyline(int[] xPoints, int[] yPoints, int nPoints)	Works like drawPolygon, but does not connect the 1st and last points
void	drawRect(int x, int y, int width, int height)	Draws a rectangle with a top left corner at (x,y) and covers the given width/height.
void	drawRoundRect(int x, int y, int width, int height, int arcWidth, int arcHeight)	Draws a rectangle with rounded corners. The corners are rounded based on arcWidth/arcHeight.
void	fillArc(int x, int y, int width, int height, int startAngle, int arcAngle)	Fills in a circular or elliptical arc covering the specified rectangle.
void	fillOval(int x, int y, int width, int height)	Fills in an oval inscribed in a rectangle with a corner at (x,y) and covers the given width/height.
void	fillPolygon(int[] xPoints, int[] yPoints, int nPoints)	Fills in a polygon with points taken from the xPoints and yPoints arrays. nPoints is the length of the arrays.
void	fillPolygon(Polygon p)	Fills in polygon p.
void	fillRect(int x, int y, int width, int height)	Fills in a rectangle with a top left corner at (x,y) and covers the given width/height.
void	fillRoundRect(int x, int y, int width, int height, int arcWidth, int arcHeight)	Fills in a rectangle with rounded corners. The corners are rounded based on arcWidth/arcHeight.
void	setFont(Font font)	Sets the font to the given font
void	drawString(String str, int x, int y)	Draws the text at (x,y)

Changing the Drawing Color

Method 1 Format:

graphicsObject.setColor(Color.COLOR);

Note: The available colors can be found in the documentation of Color.

Example:

```
g.setColor(Color.RED);        // changes the drawing color to red
```

Method 2 Format:

graphicsObject.setColor(new Color(red, green, blue));

Note: red, green, blue are each int values from 0 to 255

Example:

```
g.setColor(new Color(0,213,230));    // changes the drawing color to teal
```

Setting Up a Graphics Project

1. Create a mainFile
   a. In the main method create an instance of the JFrame
2. Create a JFrame with the following code:

```
import java.awt.*;
import javax.swing.*;

public class MyFrame extends JFrame
{
        public FrameFile_SPS(String frameName)
        {
                super(frameName);
                setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
                setResizable(false);
                pack();

                MyPanel p = new MyPanel ();
                Insets frameInsets = getInsets();

                int frameWidth = p.getWidth()
                        + (frameInsets.left + frameInsets.right);
                int frameHeight = p.getHeight()
                        + (frameInsets.top + frameInsets.bottom);

                setPreferredSize(new Dimension(frameWidth, frameHeight));

                setLayout(null);
                add(p);
                pack();
                setVisible(true);
        }
}
```

3. Create a JPanel with the following Code:

```
import java.awt.*;
import javax.swing.*;

public class MyPanel extends JPanel
{
        public MyPanel ()
        {
                super();
                setSize(400,400);          // width,height
}

public void paint(Graphics g)
{
        // code to draw to the screen using Graphics Commands
}
}
```

Terms

Frame Borders	The thin bars that frame out a window.
Resolution	The number of pixels being display horizontally and vertically.
Title Bar	The bar at the top of a frame that has the title of the window, the minimize button, the maximize button and the close button.